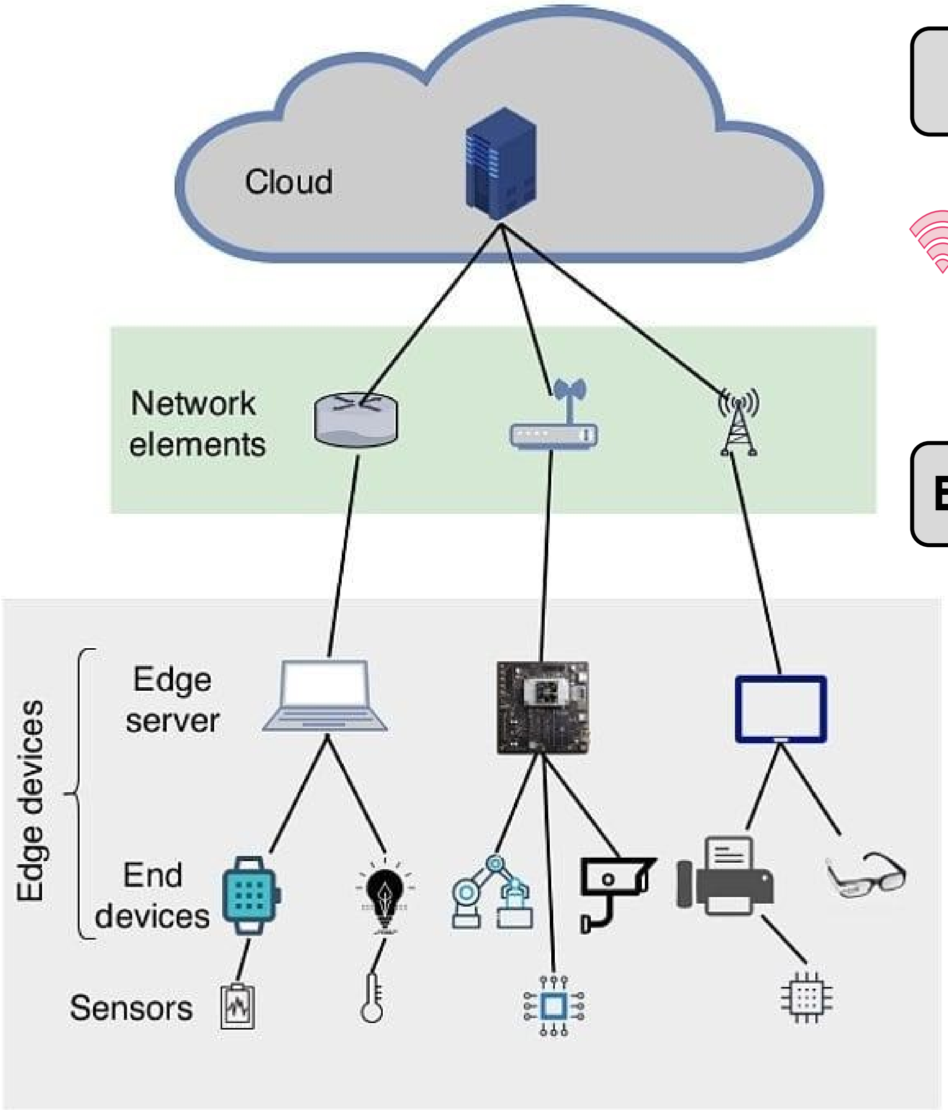
Introduction to Edge AI

From Cloud Intelligence to Device Intelligence

- What is Edge AI?
- Traditional Cloud AI Vs Edge AI
- Why Edge AI is Important
- Real-World Use Cases
- Edge AI Architecture
- Edge + Cloud = Hybrid AI
- Challenges of Edge AI
- Transition to TinyML

What is Edge AI?



The "Edge" = Where data is generated

- Sensors
- IoT devices
- Embedded systems
- Smart phones

Edge AI = AI processing directly on devices (at the edge)

- Smart cameras
- Wearables
- Industrial sensors

Edge AI enables real-time intelligence at the source

⚡ Moves intelligence closer to data sources
🤖 Enables local real time decision-making
✓ Reduces delay & network dependence

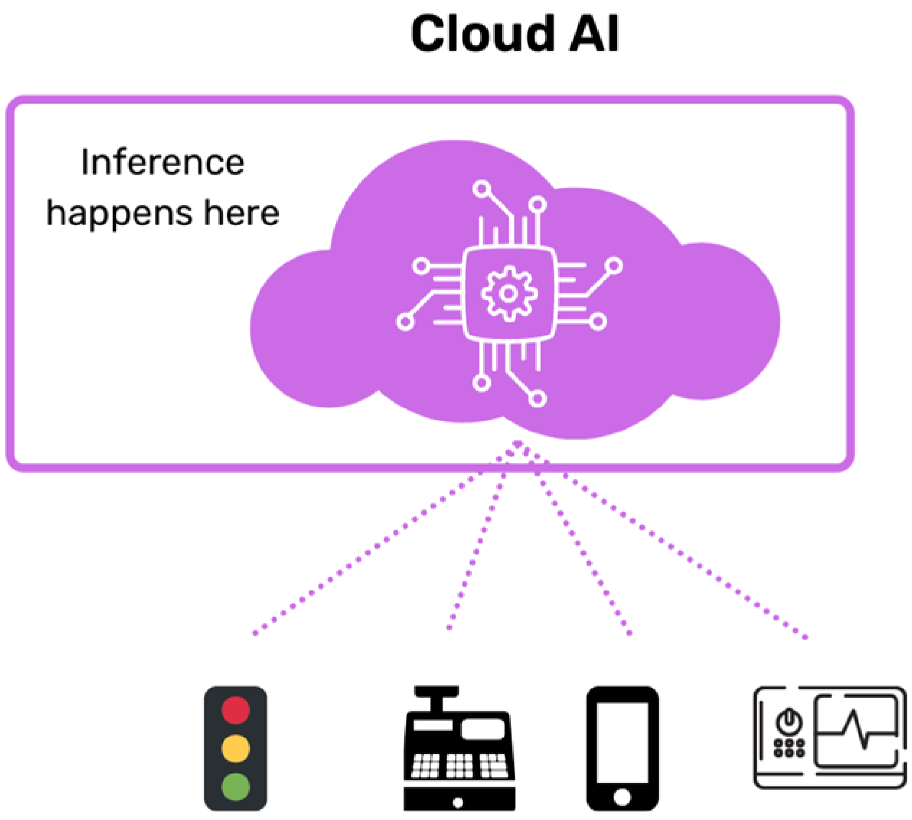
Why Edge AI?

Traditional Cloud-Centric Processing

- ☁ Devices send data to the cloud for analysis
- ⚠ Latency delays decision-making
- 📡 Depends on network connectivity
- 💰 High bandwidth & transmission costs

Limitations of Cloud-Only AI

- ⏱ Slower response times
- 📶 Connectivity constraints in remote environments
- 📈 Massive data transfer overhead



Why Edge AI?

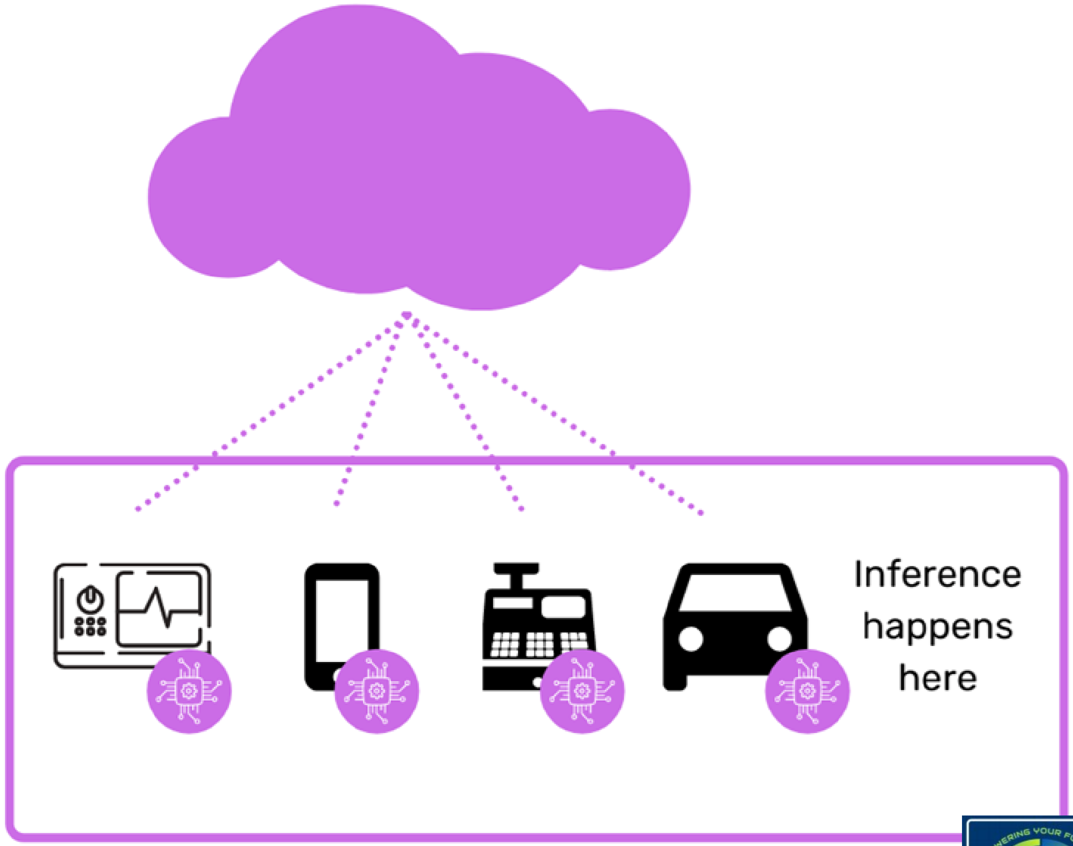
Edge AI: A Paradigm Shift

⚡ Moves intelligence closer to data sources
🤖 Enables local decision-making
✓ Reduces delay & network dependence

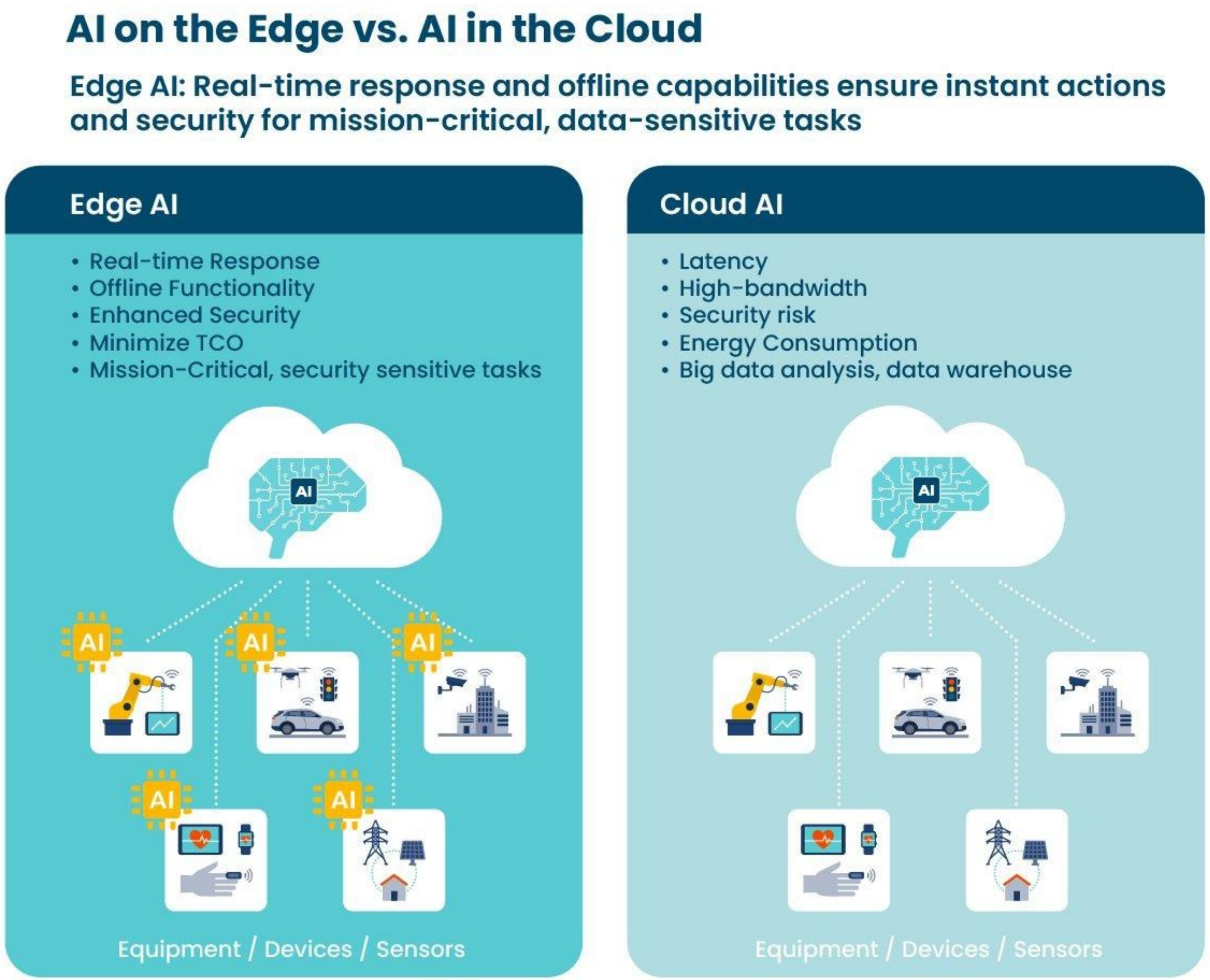
How it works?

🤖 AI models runs directly on sensors, cameras & gateways
📍 Processing occurs near the data source

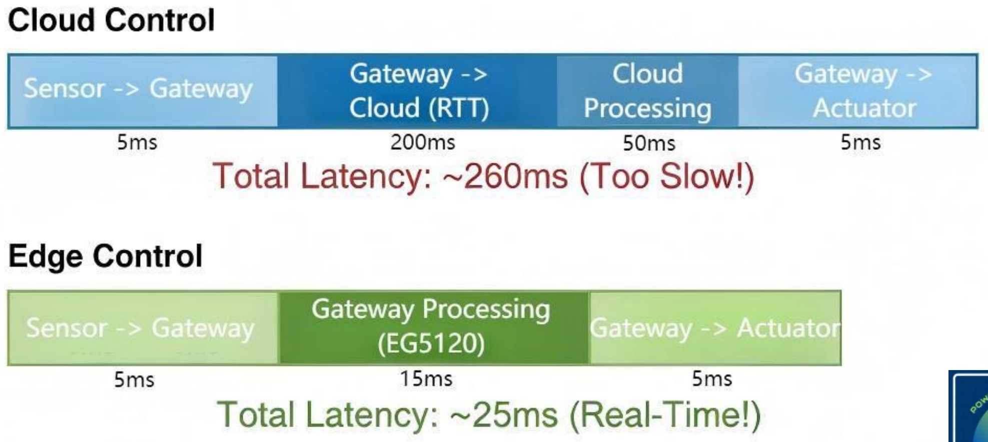
Edge AI enables real-time intelligence at the source



Edge AI vs Cloud AI



Feature	Cloud AI	Edge AI
Latency	High	Very Low
Internet	Required	Not Required
Privacy	Lower	Higher
Cost	Ongoing	One-time



Key Advantages of Edge AI

⚡ Reduced Latency

- Data processed in milliseconds on-device. Enables instant response for safety & automation

🔐 Enhanced Privacy & Security

- Sensitive data stays local. Reduces interception risks & supports compliance (e.g., GDPR)

📉 Lower Bandwidth Costs

- Only insights sent to the cloud. Minimizes data transfer & communication costs

🔄 Increased Reliability

- Devices operate independently. Continue functioning during connectivity loss

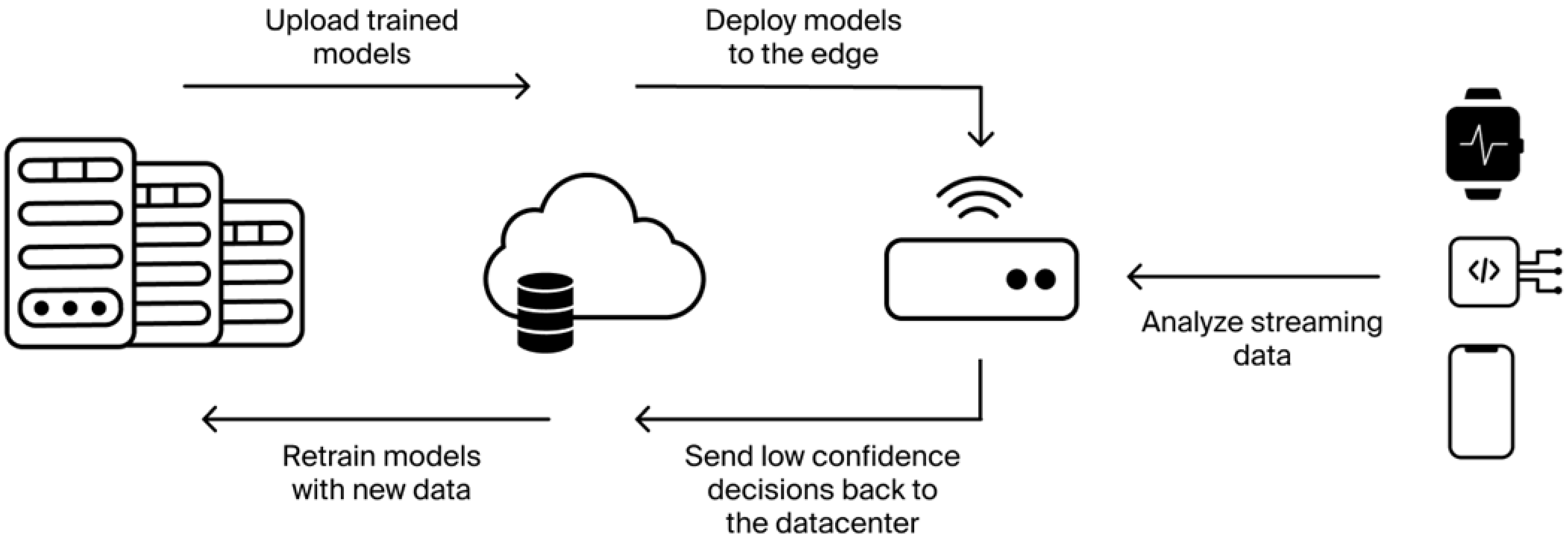
Real-World Use Cases

Critical Real-Time Applications

- Smart Surveillance
- Autonomous Vehicles
- Predictive Maintenance
- Healthcare Wearables
- Real-time anomaly detection

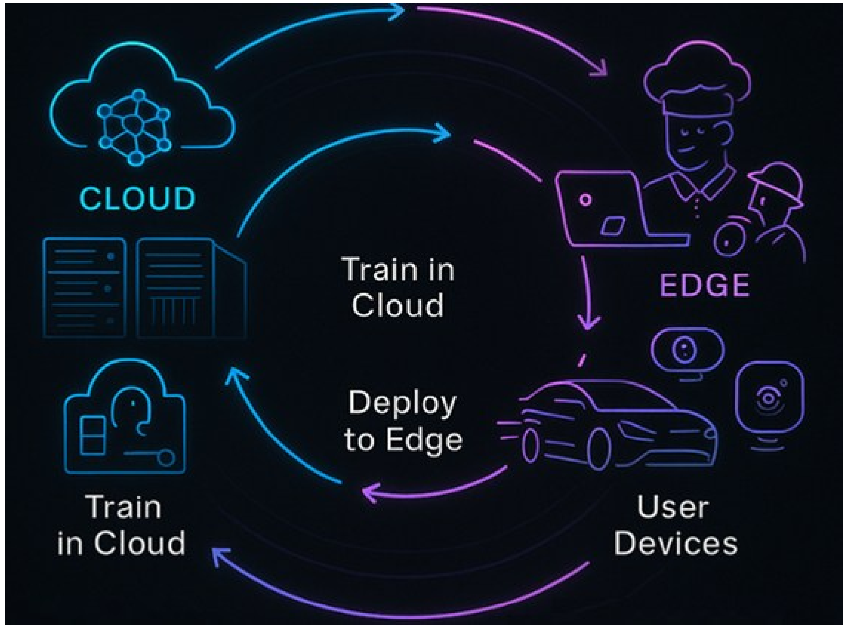
Edge AI Architecture

Data Collection | Preprocessing | Model Inference | Action / Decision



Edge + Cloud = Hybrid AI

Hybrid AI, combining edge computing and cloud resources, offers a powerful, efficient, and secure approach by processing data locally for immediate, low-latency insights while leveraging the cloud for extensive training, storage, and complex analysis. It optimizes bandwidth usage and provides privacy-by-design for applications like autonomous systems and industrial automation.



Key Components of Hybrid AI

Edge AI (Local Nodes): Processes data directly on devices for real-time decisions, reducing latency and reliance on consistent connectivity.

Cloud AI (Centralized System): Handles large-scale data aggregation, model training, and long-term analysis, requiring high computation& scalability.

Hybrid Model: Ensures data flows between edge & cloud to continuously improve algorithms. Data is preprocessed at edge, while massive retraining occurs in the cloud.

Autonomous Driving: Edge AI handles immediate obstacle detection, while cloud AI improves navigation models.

Manufacturing: Predictive maintenance and visual inspection use edge cameras for quick detection and the cloud for in-depth analysis.

Smart Homes: Local devices handle immediate user commands, while the cloud processes data for broader, long-term trends

Challenges of Edge AI

Edge AI challenges center on balancing high-performance AI models within strict resource constraints—limited power, memory, and compute—on local devices.

Limited memory

Hardware Constraints

Power and Thermal Management

Operational Management

Model Optimization & Accuracy

Security and Privacy

Common Solutions & Mitigation Techniques

- **Model Optimization:** Using techniques like quantization and pruning to fit models within limited storage and compute capacity.
- **Federated Learning:** Training models across multiple devices without exchanging sensitive, raw data, enhancing privacy.
- **Specialized Hardware:** Utilizing specialized AI chips (e.g., MCUs, ASICs, FPGAs) designed for efficient edge inference.
- **Efficient Data Management:** Implementing edge-specific databases to filter, sanitize, and manage data locally.
- **Enhanced Security Protocols:** Deploying robust, regular security updates and encrypted, secure communication channels.

Edge AI – Critical Components & Emerging Trends

Advances in hardware & software are making Edge AI practical and scalable

Hardware Accelerators

- GPUs • VPUs • NPUs enable on-device AI processing
- Run complex models efficiently at the edge

Power-Efficient AI Processing

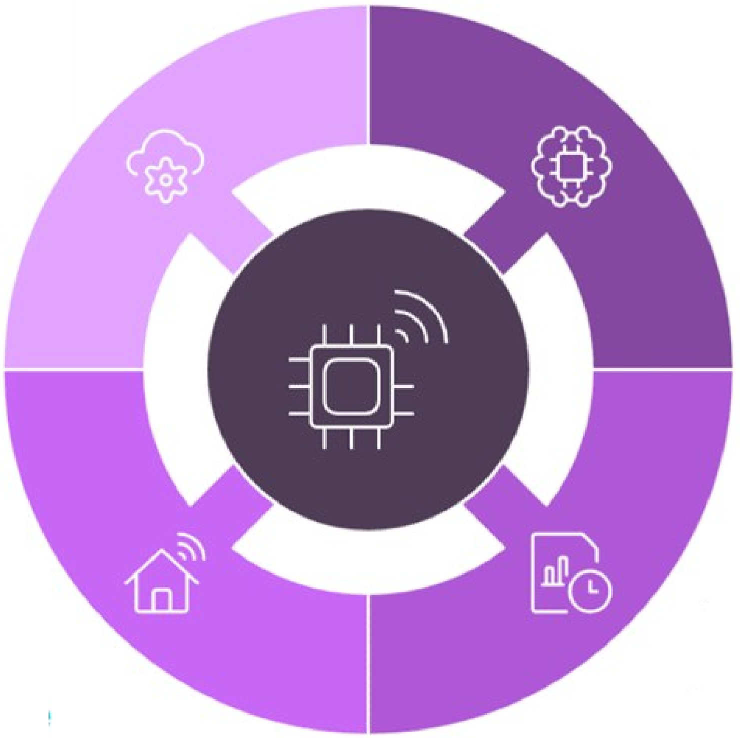
- Ultra-low power AI chips for battery-operated devices
- Enables smart wearables & remote sensors

Edge AI Platforms

- NVIDIA Jetson (AGX Orin)
- MediaTek Genio platforms
- Provide high-performance edge computing

🛠 Software & Development Ecosystem

- Intel OpenVINO • Edge Impulse
- Optimize, deploy & manage edge AI models



Hardware Acceleration Libraries & Platforms

High-Performance Edge AI Platforms

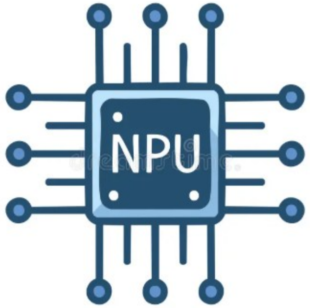
NVIDIA Jetson + TensorRT & DeepStream

✓ GPU acceleration for video analytics & robotics
✓ Optimized inference & real-time processing

Low-Power AI Acceleration

Google Edge TPU / Coral

✓ ASIC designed for TensorFlow Lite models
✓ High-speed inference with minimal power usage



Embedded Vision & Industrial AI

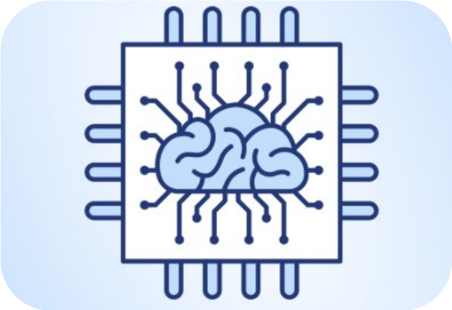
Qualcomm Neural Processing SDK

✓ Optimizes AI on Qualcomm chipsets
✓ Ideal for smart cameras, drones & IIoT

Microcontroller AI Deployment

STMicroelectronics X-CUBE-AI / Edge AI Suite

✓ Converts models into optimized C code
✓ Enables AI on STM32 microcontrollers



Autonomous Edge Intelligence

NXP eIQ Agentic AI Framework

✓ Enables real-time autonomous decision-making
✓ Optimized for NXP edge processors

Hardware acceleration enables powerful, efficient AI inference at the edge

Edge AI Deployment & Management Tools (MLOps)

MLOps (Machine Learning Operations) is a set of practices that combines machine learning, DevOps, and data engineering to automate, deploy, and maintain machine learning models in production efficiently.

AWS IoT Greengrass / SageMaker Edge Manager: Provides cloud-native deployment, monitoring, and secure updates for models running on edge devices.

Azure IoT Edge: Distributes cloud-based AI functionality (trained in Azure ML) to physical edge devices.

Balena: A platform for managing containerized applications across device fleets, ideal for, deploying AI models in real-time.

Mender: An open-source software solution for secure and reliable OTA updates for IoT devices and Edge AI models.

Foundries.io: Focused on secure, Linux-based software development and deployment for embedded devices.

MLOps tools enable secure, scalable, and remote edge AI operations

Top Edge AI Frameworks & Libraries

⚙ TinyML & Embedded AI

TensorFlow Lite / TFLite Micro

✓ Deploy AI on microcontrollers & embedded devices
✓ Supports quantization & pruning for low memory use

🔄 Cross-Platform Inference

ONNX Runtime

✓ Run models across diverse hardware platforms
✓ High-performance, optimized inference

📱 Mobile AI Deployment

PyTorch Mobile

✓ Deploy models on iOS & Android
✓ Hardware acceleration support

🚀 Model Optimization & Compilation

Apache TVM

✓ Optimizes models for CPUs, GPUs & custom accelerators

🧠 TinyML Development Platform

Edge Impulse

✓ Build & deploy low-power models for sensors & microcontrollers

👁 Computer Vision Processing

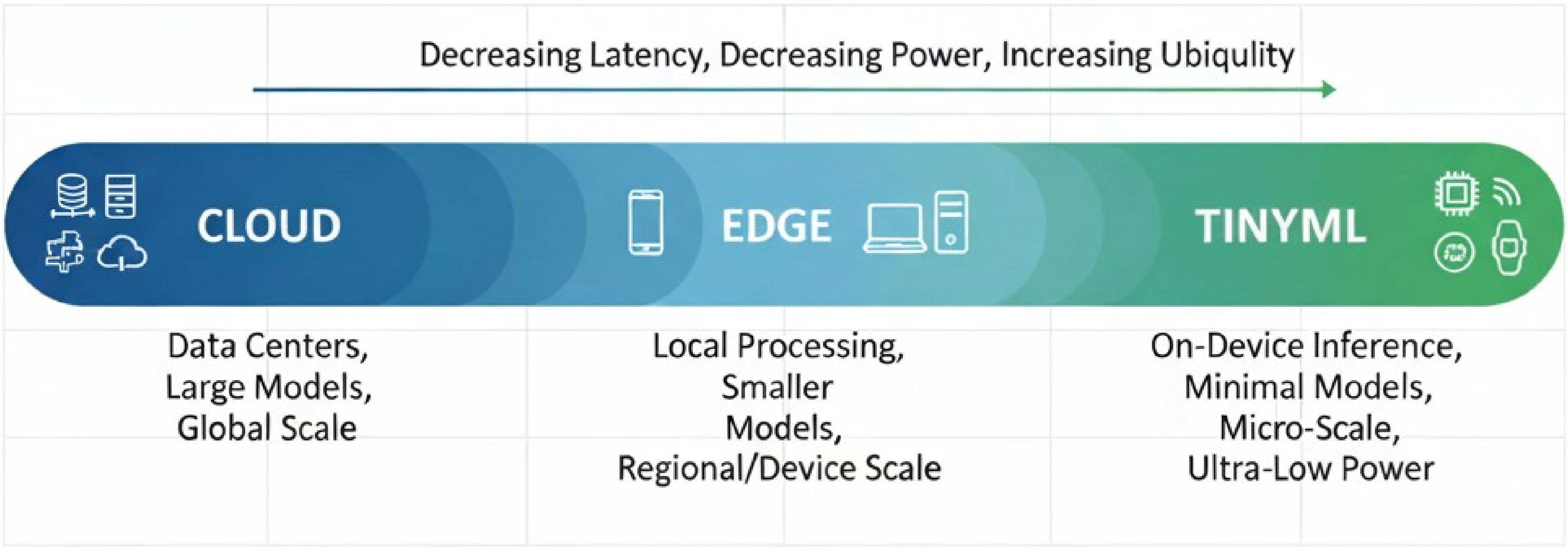
OpenCV

✓ Real-time image & video processing on edge devices

These frameworks make AI efficient, portable, and edge-ready

Transition to TinyML

From Edge AI → TinyML



TinyML

- Running AI on microcontrollers
- Ultra-low power devices
- KB-level memory

Edge AI : Key Takeaways

From Cloud to Edge Intelligence
👉 Shift from centralized → distributed AI
👉 Intelligence moves closer to data sources

🔹 What is Edge AI?
- AI processing on devices (edge)
- Real-time, low-latency decision-making

🔹 Why Edge AI?
- Instant insights
- Reduced latency & bandwidth
- Improved privacy & reliability

🔹 Edge AI vs Cloud AI
- Edge → Fast, local, efficient
- Cloud → Powerful, scalable, analytical
- Together → **Hybrid AI**

🔹 Architecture Overview
- Data → Preprocessing → Inference → Action
- Continuous feedback loop with cloud

🔹 Real-World Applications
- Smart surveillance
- Autonomous vehicles
- Predictive maintenance
- Healthcare wearables

🔹 Key Enablers
- Hardware accelerators (GPU, NPU)
- Edge platforms & SDKs
- AI frameworks (TFLite, ONNX, OpenCV)
- MLOps for deployment & management

🔹 Challenges & Solutions
- Constraints: memory, compute, power
- Solutions: optimization, specialized hardware, federated learning

🔹 The Big Picture
👉 Edge AI = **Real-time intelligence at the source**
👉 Foundation for **TinyML & next-gen AI systems**